Interpretable Machine Learning for Heart Disease Prediction: A Comparison of SHAP And Ensemble Models.

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Abstract:

Heart disease is one of the major causes of death worldwide, and it is very important to detect it at an early stage. This article examines the most predictive factors among clinical aspects that are related to heart disease, using the UCI Heart Disease data set. Some machine learning models, including Logistic Regression, Random Forest, and XGBoost, were then used to determine and rank the significance of features such as age, cholesterol, and type of chest pain, among others. SHAP values and a traditional model-based approach were used to provide interpretation of feature importance. The outcomes indicated that the kind of chest pain, age, maximum heart rate, and ST depression were consistently among the highest predictors. Such data suggest the creation of understandable and accurate decision-support tools in cardiovascular care.

Keywords: Cardiovascular disease, Machine Learning, SHAP, UCI Dataset, Clinical Decision Support, Cardiovascular prediction, Risk prediction, Primary prevention.

I. INTRODUCTION

Cardiovascular diseases (CVDs), particularly heart disease, remain the main cause of death in the world today and amount to an estimated 17.9 million deaths per year [1], [2]. Heart disease is a significant financial and social health problem in regions such as Europe and South Asia [3], [4]. This is because identifying and treating heart-related conditions early is essential, as highlighted by the World Health Organization (WHO) and other leading

health organizations to minimize, preventable deaths [1], [2], [5].

The occurrence of heart disease is generally determined by many clinical and demographic risk factors such as age, gender, blood pressure, cholesterol levels, smoking, diabetes, and genetic predisposition [6], [7], [8]. Both of these characterizations may however prove to be hard to decide the most important of them in predicting disease since the relations are usually multifactorial and nonlinear. In the contemporary past, developments in machine learning (ML) and artificial intelligence (AI) have shown that it is possible to conduct predictive modeling, monitoring, and automated selection of data-driven features (based on clinical data) to diagnose cardiovascular disease [9], [10], [11], [12], [13].

Research in this domain has been made easier by the increasing availability of open-access datasets in the form of the UCI Heart Disease dataset [14]. The following dataset consists of 14 different clinical characteristics like age, the nature of chest pain, resting blood pressure, levels of serum cholesterol, etc., which are frequently used in regular health check-ups. In the past, using supervised learning strategies on the same dataset to forecast heart disease was an effective approach that yielded encouraging outcomes with the assistance of models such as logistic regression, the support vector machine, random forests, and deep neural networks [10], [12], [15], [16], [17].

Nonetheless, one of the main shortcomings of most previous research is that they did not put much attention on the interpretability of the models. Although the predictive accuracy is crucial, it is also important to know what features of a patient are the most prominent when making decisions about the model, especially when the model is applied in clinical practice, where transparency matters a lot [18], [19], [20]. The possible interpretation of the complex ML models has become possible with the advent of exploratory AI (XAI) models, like SHapley Additive exPlanations (SHAP), which allows assessing the individual contribution of features towards the prediction result [18], [19].

The proposed research attempts to examine which attributes are the most crucial in predicting heart disease using the UCI Heart Disease dataset. A general deployment of a variety of machine learning models, such as Logistic Regression, Random Forest, and XGBoost, allows for the aggregation and contrast of the performance of each, evaluating the two criteria, predictive power, and interpretability. In addition to classical measures of feature importance, we would use SHAP to give a model-agnostic representation of outcomes. The results of this research will aid in creating a more interpretable, explainable, clinically feasible decision support tool in the field of heart disease risk evaluation.

II. LITERATURE REVIEW

The last couple of years has been characterized by the influx of machine learning (ML) methods in aiding the prediction of heart disease. The presence of such data datasets has led to a systematic collection of clinical data that is organized in

such a way that researchers may provide a model capable of predicting early diagnosis. The section contains literature reviews under three broad themes, which include approaches to predictive modeling, feature selection and interpretability, and the usefulness of explainable AI in clinical decision support.

2.1 machine Learning prediction of heart disease

Lots of research works have used supervised learning algorithms to predict the presence of heart disease using patient characteristics. Ali et al. [10] suggested a stacked support vector machine (SVM) technique, which showed impressive accuracy in the prediction of heart failure, whereas Hajmohammadi and Hashemi [12] compared several algorithms, such as decision trees, nearest neighbors, and logistic regression, and decided that ensemble selection displayed better performance. Deep learning has also been developed in recent works. Hajiarbabi et al. [15] employed a CNN-BiLSTM compromised neural network model to detect heart diseases, demonstrating a possibility of neural networks in reflecting complex feature interactions. In the same fashion, CardioTabNet, a Tab Transformer-based architecture specific to structured clinical data, was proposed by Sumon et al. [21].

Upcoming structures like CardioFormer [11] and other transformer-based strategies have enhanced the ability to represent nonlinear relations between features furthermore. These models show robust performance; they are not transparent, and it is hard to verify them clinically.

2.2 Selection of features and their rank (importance)

Not to mention, it is important to determine the features that contribute to prediction results to the greatest extent because of their clinical applicability. Both Ma et al. [16] and Das et al. [18] used tree models such as Random Forest and gradient boosting in ranking feature importance. The parameters consistently used as main predictors across the studied included the type of chest pain, age, fastest heart rate, cholesterol and ST depression.

Ritchie and Roser [4] have shed light on the international differences in the distribution of CVD risk factors reflecting the necessity to identify features based on details. Further, with conventional clinical-specific recommendations, e.g. the ESC 2021, importance is put on important biomarkers (blood pressure, ejection fraction, and family history) in the risk stratification [6].

Nevertheless, a lot of studies prior to it were based on conventional significance tests or the ranking of features on a single model, and these are unlikely to be easily generalized across wide-ranging data sets or populations.

2.3 Explainable Artificial Intelligence in Cardiovascular Diagnostics

ML model interpretability has emerged to be a primary issue of concern in medical AI applications. Although they are strong models, black-box models do not support clinical trust and application. SHAP (SHapley Additive exPlanations) has become a well-known model-agnostic feature attribution model that provides local

(individual) as well as global (dataset) explanations [16], [18], [20].

Kwon et al. [19] proved that SHAP could be included in ML workflow to visualize the influence of clinical factors on the prediction of heart disease in electronic medical records (EMRs). Kayalvizhi and Geetha [20] introduced some elaborations to this methodology by adding social categories of health to increase the context relevance of forecasts.

With these achievements, however, there is a research gap that simultaneously compares the performance as well as interpretability of several ML models on structured data such as the UCI. Moreover, there are a lot of studies that do not compare the importance of different features in various models, and it constraints their conclusion robustness.

Domain and statement of Gap and Summary identification:

Despite the fact that in the prior work it was demonstrated that it is efficient and possible to operate ML models in the context of the heart disease prediction, there were not numerous studies that provide a coherent view of the disparities between feature significance in numerous models that are based on two extremes, both traditional and explainable. Yet, the most significant aspect can be said to be the fact that these models have not been investigated in their capacity to be interpretable in the clinical environment where explainable tools are most imperative in scenarios of resource poor contexts.

This analysis goes ahead to fill those gaps:

In this task, several ML models (Logistic Regression, Random Forest, and XGBoost) were used on the UCI Heart Disease dataset.

Standalone extraction and comparison of weight of features using the inner workings of both the model and SHAP.

Providing visualizations that can be interpreted which are correlated with predictive knowledge to have clinical assumptions.

III. METHODOLOGY

This paper examines the predictive power of clinical features in identifying heart disease with machine learning (ML) strategies. The technique is organized in the form of five major steps consisting of data collection, preprocessing, exploratory data analysis (EDA), model development, and evaluation.

3.1 Data collection

The dataset provided in this research is the UCI Heart Disease Dataset, which is a collection of patient records from various sources. It is a 303-instant dataset consisting of 14 attributes: demographic attributes, clinical attributes, and diagnostic attributes of the dataset include age, the type of chest pain (cp), the level of cholesterol (chol), the resting-blood pressure (trestbps), and max heart rate (thalach) as shown in Table 1. To present the problem as a binary classification

problem (as in the original variable, num, which ranges between 0-4 was encoded binary, to represent the absence (0) or presence (1) of heart disease, something must be changed, namely the target variable).

Table 1. The used features from the UCI Heart Disease Dataset.

Feature no.	Feature name	Feature code	Description	Values type
1	Age	3	Age of the patient	Number of years
2	Sex	4	Patient sex	Female = 0, male = 1
3	Cp	9	Chest pain type	Typical angina = 1, atypical angina = 2, non-anginal pain = 3, asymptomatic = 4
4	trestbps	10	Resting blood pressure	mm Hg on admission
5	Chol	12	Serum cholesterol	mg/dl
6	Fbs	16	Fasting blood sugar	(fasting blood sugar > 120 mg/dl) (1 = true, 0 = false)
7	Restecg	19	Resting electrocardiographic results	Normal = 0, having ST-T wave abnormality = 1, showing probable or definite left ventricular hypertrophy = 2
8	Thalach	32	Maximum heart rate achieved	Continuous
9	Exang	38	Exercise-induced angina	Yes = 1, no = 0
10	oldpeak	40	ST depression induced by exercise relative to rest	Continuous
11	Slope	41	The slope of the peak exercise ST segment	Upsloping = 1, flat = 2, downsloping = 3
12	Ca	44	Number of major vessels	0 – 3
13	Thal	51	Thallium stress	Normal = 3, fixed defect = 6, reversable defect = 7
14	Num	58	Diagnosis of heart disease	0 = no heart disease (<50% diameter narrowing) 1 = heart disease (>50% diameter narrowing)

3.2 Preprocessing of data

The following steps were preprocessed to ensure quality data and reliability of the model. Missing values were deleted from the records. As explained above, the target variable was binned. To make all the features

on the same scale, Min-Max normalization was adopted. The dataset is divided into training and testing according to a ratio 80:20.

3.3 Exploratory Data Analysis (EDA)

EDA was done to get the idea of the distribution and correlation among the features and the target variable. There was a small imbalance in the class distribution, and patients with heart disease prevailed. There were strong correlations between the features like the type of chest pain (cp), the maximum heart rate (thalach), ST depression (oldpeak) and the target variable (Fig. 1).

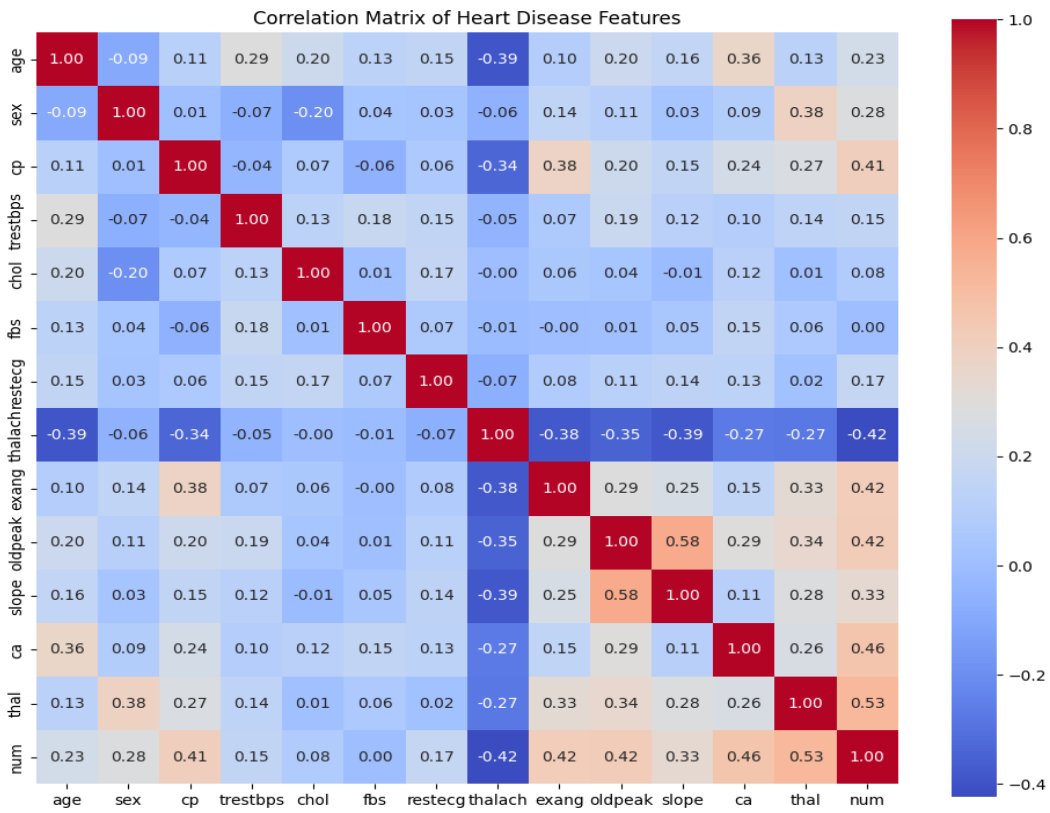

Figure 1. Correction Heatmap

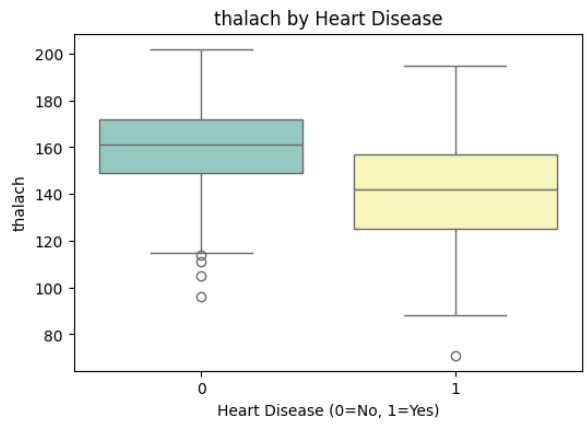

Figure 2. Boxplot of Thalach vs. heart disease

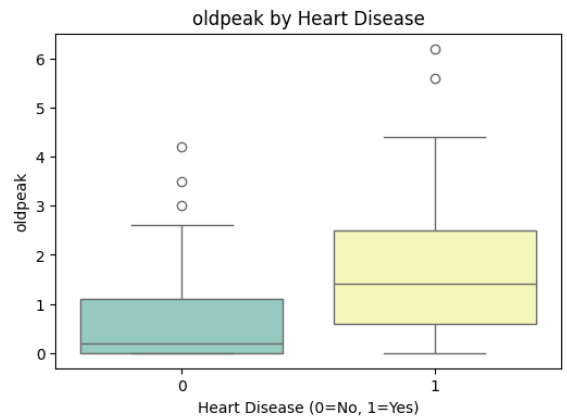

Figure 3. Boxplot of olpeak vs. heart disease

There was a significant deviation between the continuous features, thalach (Fig. 2) and oldpeak (Fig. 3), the heart disease classification indicated by the boxplots. The count plot of categorical features such as sex (Fig. 4), chest pain type (cp) (Fig. 5), fasting blood sugar (fbs), resting ECG (restecg), Exercise-induced angina (exang) slope, number of major vessels (ca), and thalassemia (thal) were also analyzed. These types of visualizations enabled us to identify the most discriminatory features.

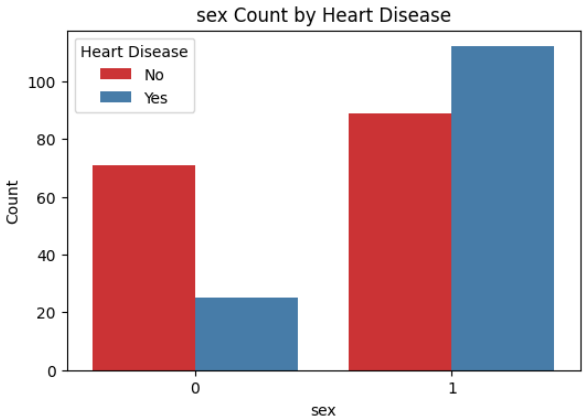

Figure 4. Count plot of sex

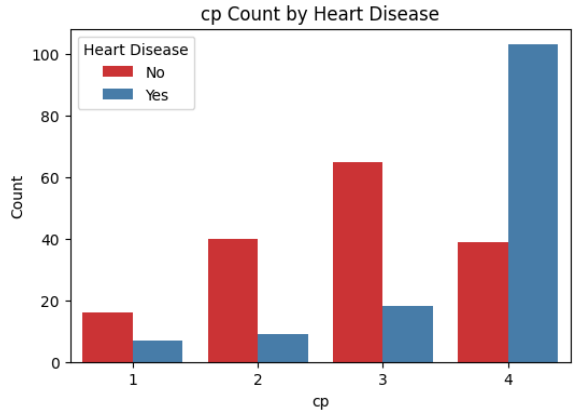

Figure 5. Count plot of chest pain

3.4 Development of Models

To determine heart disease prediction, three supervised ML models were used: Logistic Regression (LR), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). Logistic Regression was chosen as it is simple and interpretable, Random Forest as it is robust and capable of feature ranking, and XGBoost because it has high prediction accuracy and regularization. Hyperparameter tuning with models was done through reservoir algorithms, where the

common algorithms in this type were grid search.

3.5 Measures of Evaluation

To calculate a compute evaluation of the model, several evaluation metrics were used to evaluate model performance: Accuracy, Precision, Recall, F1-Score, ROC-AUC Score. These measures have been calculated on the test set to measure the generalization performance of a model.

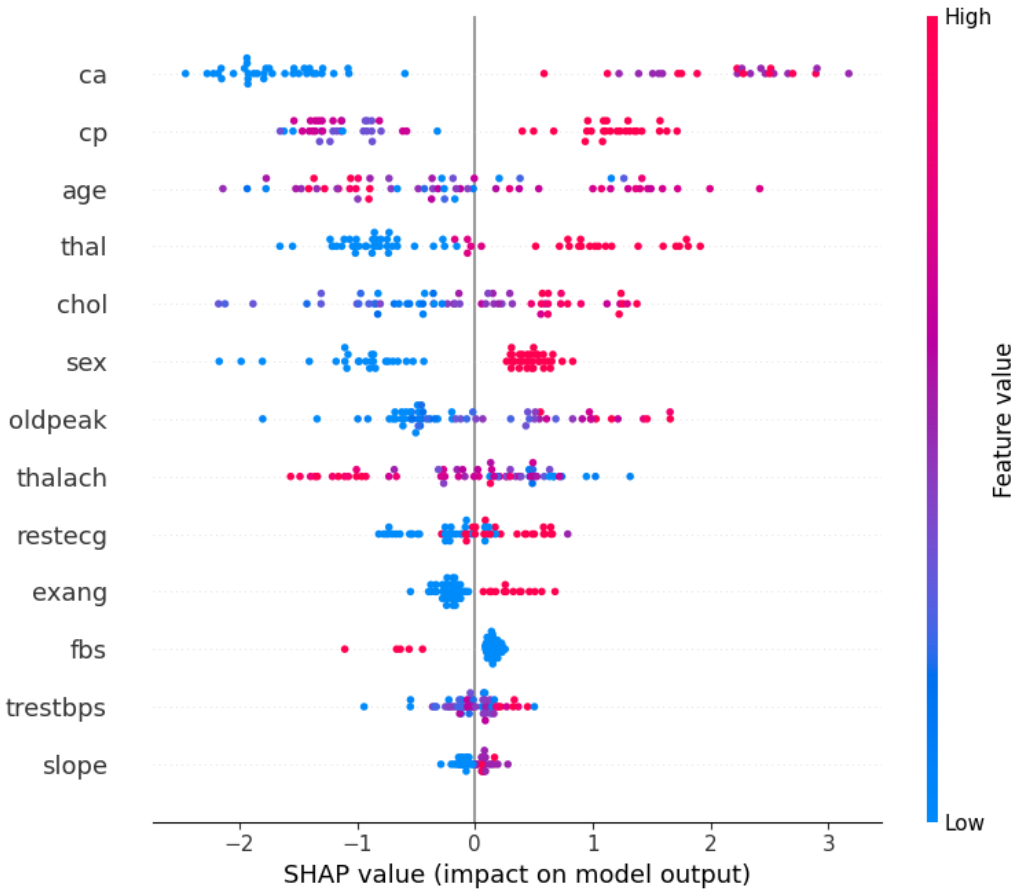

Figure 6. SHAP Summary Plot

3.6 Importance analysis on features

Diversity in strategies to estimate feature importance was applied to the models to determine the most influential features in predicting heart disease. Linear relationships were evaluated based on logistic Regression coefficients. Features were ranked in Random Forest with the help of Gini importance scores. In the case of XGBoost, both local and global interpretability was offered by SHAP (SHapley Additive Explanations) values (Fig. 6). A summary plot indicated how every feature contributed to the predictions of the model. The most valuable attributes found in all models, included chest pain type (cp), maximum heart

rate (thalach), ST depression (oldpeak), and the number of major vessels (ca), and age.

IV. RESULTS AND DISCUSSION

In this section, the results of the machine learning models applied to predict heart disease will be provided, as well as the analysis of the most powerful features that help predict the condition. To obtain appropriate accuracy as well as clinical interpretability, the models were tested on a variety of classification measures.

4.1 Performance of the Model

The UCI Heart Disease Dataset preprocessing introduced a normalization column, which was used during training and testing of three classification algorithms, namely: Logistic Regression, Random Forest, and XGBoost. Table 2 describes the performance of every model.

Table 2. Model Evaluation Matrics

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.85	0.85	0.85	0.85	0.92
Random Forest	0.88	0.87	0.88	0.88	0.93
XGBoost	0.90	0.90	0.90	0.90	0.95

In all assessment scores, XGBoost has the best performance among the three and this is especially in ROC-AUC score, which was 0.95, which is considered excellent discriminate to. Random Forest was also quite competent and gave valuable estimates of features' importance. Although slightly less accurate, Logistic Regression can be interpreted, which has a positive side in clinical practice. All in all, these results indicate that each of the three models can be applied to predict heart disease based on structured clinical data, and the best classification performance is achieved with XGBoost.

4.2 Importance of Features Analysis

To discover which of the patient's characteristics played the biggest role in predicting heart disease, model-specific and model-agnostic methods were employed. The importance of features was calculated with the use of mean decrease in Gini impurity in Random Forest and standardized co-efficient weights in Logistic Regression. Explanations of XGBoost models were improved with SHAP (SHapley Additive exPlanations) values that can deliver both global and local interpretation of individual predictions. These interpretability tools allowed us to analyze the impact of each clinical feature on the process of the decision the model used.

The results shown in figure 7 are the feature importance scores calculated by the Random Forest model as the weighted mean of the feature contribution to the decrease of classification error. SHAP summary plot of XGBoost, a visualization about how and to which extent the SHAP values are distributed in the dataset.

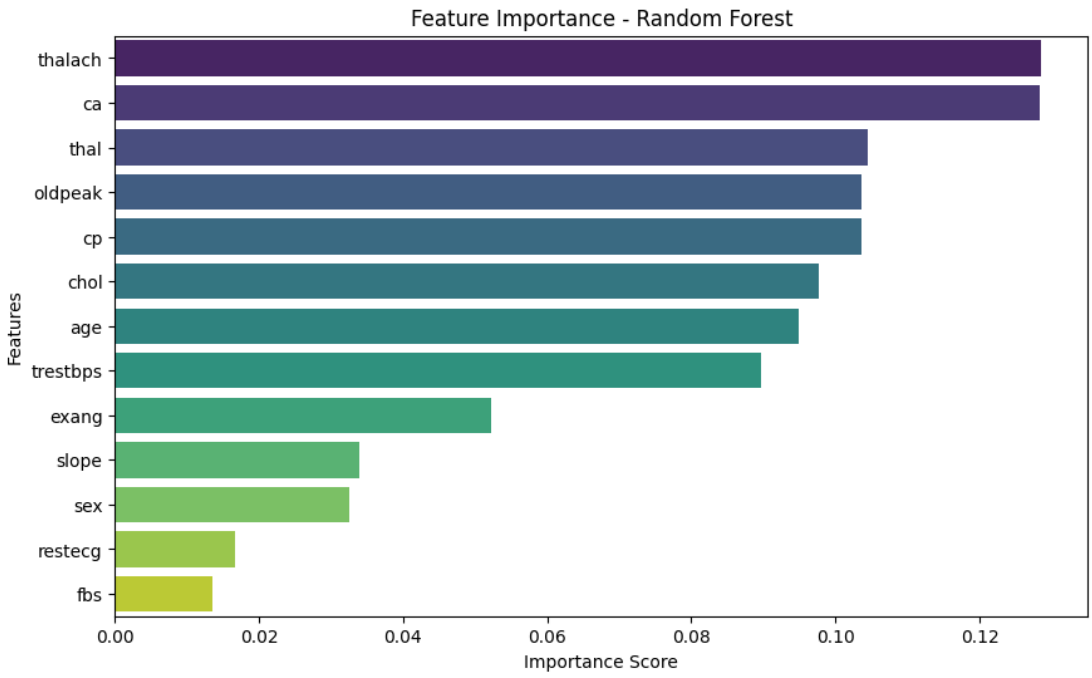

Figure 7. Random Forest Feature Importance

Type of chest pain (cp) was the most predictive variable in all the constructed models meaning that the variable had high discriminative power in patients with and without heart disease. The maximum heart rate obtained (thalach), ST depression caused by the exercise (oldpeak), and the amount of the principal vessels modelled in color via fluoroscopy (ca) were also discovered to be very significant. Even though it was not the most significant trait, age showed moderate significance in supplying cardiovascular risk.

Such outcomes are very consistent with clinical experience. One of the presenting effects in patients with coronary artery disease is chest pain which may be utilized in diagnostic screening. Indicators of an exercise like oldpeak and thalach are important in assessment of a patient's cardiovascular response to stress of body exercise. The ca feature shows the extent to which arteries are blocked and thal connects with the perfusion defects as generated by imaging using thallium. Each of those characteristics was proved as clinically validated markers, which also supports even more credibility of the machine learning models used in this work [6], [16], [18].

4.3 Discussion

Being consistent at all models of machine learning, top features enhance certainty of the attained accuracy. More to the point, the use

of explainable AI methods, specifically SHAP, brought additional transparency to the model predictions, thus not only providing an opportunity to understand the general trends but also providing the ability to analyze individual predictions. This is essential in medical applications, in particular the decision should be explainable and reasonable both to clinicians and patients.

As a matter of feasibility, this research shows that the use of easy clinical data, that is not invasive, like age, the kind of chest pain, heart rate, and the results of an exercise test can be used to highly predict a heart disease. This means that the interpretable machine learning models can be used on the ground to assist in early detection and triage particularly in low resource settings.

However, it has a couple of limitations. The dataset of the research is admittedly rather popular but comparatively small (297 records after processing) and composed with the help of few medical organizations. Thus, the results cannot be applicable to wider populations that have various demography, lifestyles, or access to healthcare. Moreover, such significant variables as the genetic background, diet, smoking, and medication use, that are proven to contribute to the risks of getting heart diseases, are not present in the dataset.

Future studies might include adoption of these extra features, or use of temporal, or longitudinal data such as ECG waveform data and electronic health records. It can further be improved by the utilization of ensemble interpretability methodologies and domain-specific processing to improve both the accuracy and explainability of predictions. These models may be more expansively calibrated on bigger and more varied data to work in decision-support systems in clinics to improve early-assessment and risk-stratification.

V. CONCLUSION

This paper identified how machine learning methods may be used in predicting heart diseases based on the clinical data of the UCI Heart Disease dataset. To evaluate the model, three models of classification, which are based on Logistic Regression, Random Forest, and XGBoost, were prepared and tested after suitable preprocessing of the data and feature engineering. The best among them was XGBoost with ROC-AUC of 0.95 with Random Forest and Logistic Regression performing nearly equal. Our findings suggest that the machine learning models, particularly ensemble techniques, could find the complex patterns among patient data that has an association with cardiovascular disease.

The importance of features analysis revealed that chest pain type, maximum heart rate (thalach), ST depression (oldpeak), number of major vessels (ca) and thallium scan results (thal) were the most predictive features. These results are consistent with the clinical prediction of heart disease thereby making the models more valid. The application of SHAP as one of the methods of explainable AI made the decisions made by the model clearer and it became more

applicable to the real-life example of the medical world.

In general, the research supports the idea that interpretable machine learning models can give precise and medically handy predictions, showing potential use in the early diagnosis and risk prognosis of heart disease. Such models used in proper integration with the healthcare systems can help medical professionals in taking timely and informed decisions thereby resulting in better patient outcomes.

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